Fog Model-Based Hyperspectral Image Defogging

Xudong Kang, *Senior Member, IEEE*, Zhengyao Fei, Puhong Duan, *Student Member, IEEE*, and Shutao Li, *Fellow, IEEE*

***Abstract*— Fog in hyperspectral images severely limits the visibility of imaging scene and reduces the image contrast, which has a negative effect on the following image interpretation. Defogging methods aim at restoring a high-quality image from the degraded image. Currently, most dehazing methods mainly depend on the atmospheric scattering model in computer vision and multispectral image communities. However, when these approaches are directly used to remove the fog from HSIs, they cannot produce satisfactory defogging performance. To alleviate this issue, we develop a novel fog model to achieve fog removal from hyperspectral images. First, a fog density map is calculated by differentiating the averaged bands falling into visible and infrared spectral ranges. Then, haze abundance in different spectral bands is estimated based on the pixel reflectance between two selected pixels with different haze levels. Finally, the high-quality hyperspectral image is restored by solving the defogging model. Experiments performed on a new benchmark created by ourselves demonstrate that the proposed method obtains favorable dehazing performance in contrast to other approaches in computer vision and remote sensing fields.**

***Index Terms*— Fog intensity map, haze abundance, hyperspectral image, image defogging.**

I. Introduction

HYPERSPECTRAL sensors record hundreds of spectral channels from the visible to infrared spectrum carrying rich spectral information, which is increasingly developing a popular remote sensing technology [1]–[4]. Due to this merit, hyperspectral images have been extensively explored in all kinds of applications, such as urban management [5]–[7], environmental protect [8], [9], and agriculture monitoring [10]–[12]. However, since atmosphere particles absorb and scatter the reflected light under bad atmospheric conditions, the captured data usually suffer from low contrast and poor visibility. These effects seriously interfere with subsequent object identification and its applications for users. Hyperspectral image defogging is to restore high-quality images from the corrupted data.

In the computer vision field, a variety of defogging approaches have been designed to improve the clarity and quality of RGB images, which can be roughly divided into two aspects: atmospheric scattering models and image contrast enhancement methods. These atmospheric scattering models follow three similar steps. The first step is to calculate the transmission matrix from the foggy image. The second step is to assess global atmospheric light. The final step is to estimate the high-quality image by solving the atmospheric scattering model. For example, He *et al.* [13] proposed a dark channel prior (DCP) for image dehazing. The DCP was based on an assumption that most local patches in the haze-free outdoor images contained pixels with very low intensity in at least one spectral band. Cai *et al.* [14] introduced a convolutional neural network (CNN) to estimate the transmission map to avoid the inaccurate estimation of a single image. In addition, all kinds of image priors have been developed for image dehazing, such as atmospheric illumination prior [15], smoothed prior [16], and color attenuation prior [17].

Different from the atmospheric scattering model-based approaches, contrast enhancement-based methods aim to improve the image details so as to boost the visibility of the degraded image. For example, Tan [18] used the local contrast to approximate the density of the haze and restored the true color and visibility of the hazy image by maximizing the local contrast. Li *et al.* [19] applied histogram distribution prior to enhancing the brightness and contrast of the hazy images. Ancuti and Ancuti [20] adopted a Laplacian pyramid fusion-based scheme to boost the visibility of outdoor scenes, in which the inputs were constructed by a contrast-enhancing procedure and a white balance technique.

In the remote sensing community, a number of defogging methods were proposed. Some earlier dehazing approaches focused on image enhancement techniques for multispectral remote sensing images, including histogram match [21], retinex enhancement [22], homomorphic filtering [23], and wavelet transform [24]. For example, in [21], the histogram match algorithm was used to obtain ground reflectance data for high spatial resolution satellite imagery by matching the histogram of the haze areas and the histogram of the clean regions. In [22], a contrast enhancement scheme based on multiscale retinex was proposed to deal with haze in remote sensing images. To improve the dehazing performance, many

Manuscript received May 7, 2021; revised July 13, 2021; accepted July 26, 2021. Date of publication August 11, 2021; date of current version January 17, 2022. This work was supported in part by the Major Program of the National Natural Science Foundation of China under Grant 61890962, in part by the National Natural Science Foundation of China under Grant 61871179 and Grant 61801178, in part by the Fund of Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province under Grant 2018TP1013, in part by the Science and Technology Talents Program of Hunan Association for Science and Technology under Grant 2017TJ-Q09, in part by the Scientific Research Project of Hunan Education Department under Grant 19B105, in part by the Natural Science Foundation of Hunan Province under Grant 2019JJ50036, and in part by the Fund of Science and Technology Program of Guangdong Province under Grant 2018B010107001. *(Corresponding author: Puhong Duan.)*

The authors are with the College of Electrical and Information Engineering, Hunan University, Changsha 410082, China, and also with the Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province, Hunan University, Changsha 410082, China (e-mail: xudong_kang@163.com; zhengyao_fei@163.com; puhong_duan@hnu.edu.cn; shutao_li@hnu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2021.3101491

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:54:16 UTC from IEEE Xplore. Restrictions apply.

extended versions have also been investigated to eliminate the haze in multispectral remote sensing images [25]–[27]. In [28], a trilateral filter defogging method was developed to obtain the fog-free images by solving the fourth-order partial differential equations. In [25], a two-stage dehazing method was developed by combining the homomorphic filtering and sphere model. This method was used to remove thin cloud, thick haze, and uneven haze in multispectral images.

In recent years, some improved atmospheric scattering models and dark-object subtraction models have been applied for dehazing in multispectral data [29]–[34]. For instance, Pan *et al.* [29] developed a deformed haze imaging model (DHIM) by adding a translation term to recover clear image from haze multispectral remote sensing data. In [31], a dark channel-saturation prior was used to recover haze-free multispectral data. In [33], a virtual depth-based iterative dehazing method (IDM) was proposed, where the dehazing step was used iteratively to eliminate haze progressively until obtaining a good performance. Recently, some deep learning techniques have also been applied for dehazing of remote sensing data. In [35], multiple CNNs with residual structure were proposed for dehazing of multispectral remote sensing images, where the multiple CNNs were combined in parallel to map a regression model from the fog image to the clean image. In [36], an end-to-end network was designed to obtain the haze-free data from hazy data via the channel-attention structure and residual learning model.

Although various dehazing approaches have been proposed in computer vision and remote sensing fields, these methods cannot work well for removing haze from HSIs. This is due to the fact that these advanced methods mentioned above follow some priors and constraints that are not suitable for HSIs because of lacking generalization capability. Here, an intuitive example is shown to illustrate the limitation of several existing defogging methods (see Fig. 1). It is obvious that thick fog is still in the restored images obtained by several representative defogging methods. Moreover, serious color or spectral distortions may be caused in the defogging process.

In order to overcome this issue, we construct a foggy hyperspectral database for the first time, which is acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor and China Gaofen (GF)-5 satellite. More importantly, we propose a fog model for hyperspectral image defogging, which is composed of several steps. First, the fog intensity map is calculated according to the penetration ability of fog in different spectral channels. Second, the fog abundance is estimated based on the spectral difference between two selected pixels with different haze levels. Finally, the defogging results are obtained by solving the proposed haze removal model. Experiments carried out on the created dataset validate that the proposed method can consistently outperform other state-of-the-art dehazing approaches. To sum up, the novelties of this work are as follows.

1) A novel fog model-based defogging method of hyperspectral images is proposed. The hyperspectral image contaminated with fog is modeled as clean hyperspectral data and a foggy hyperspectral image, in which the foggy hyperspectral image is considered as multiplication of fog intensity map and fog abundance in different channels.
2) We construct a defogging dataset for hyperspectral images, called Hyperspectral Defogging dataset (HDD), which is publicly available for researchers.[1] All images are captured from the AVIRIS sensor and China GF-5 satellite. To the best of our knowledge, this dataset is the first foggy hyperspectral data, which will enable researchers to assess the defogging performance of new approaches.

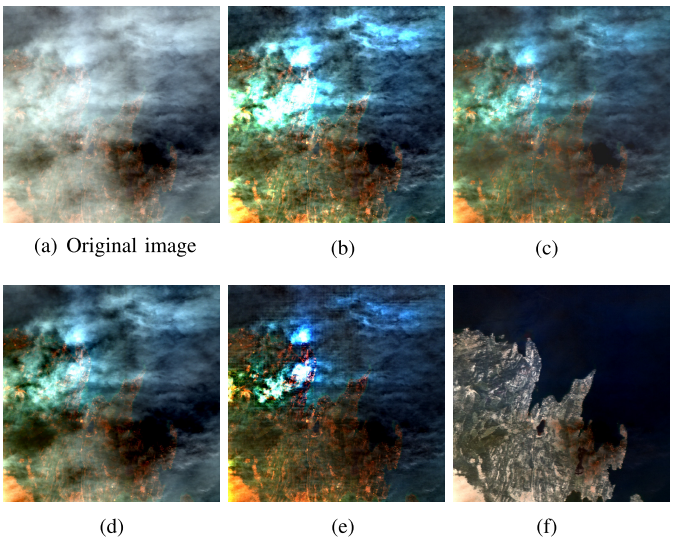


Fig. 1. Defogging results of different approaches. (a) Foggy image. (b) DCP [13]. (c) DHIM [29]. (d) MSCNN [37]. (e) MLP [38]. (f) Our method.

The rest of this work is organized as follows. Section II briefly describes the related work. Section III gives the proposed methodology. Section IV is devoted to the experiments and results. Finally, Section V presents the conclusions of this work.

II. Related Work

In recent years, various model-driven methods have been investigated for multispectral image dehazing, which can be broadly classified into two categories, i.e., atmospheric scattering model [30]–[33] and dark-object subtraction model [26], [34], [39]–[41].

First, the atmospheric scattering model is shown as

$$\mathbf{I}(x) = \mathbf{O}(x)t(x) + \mathbf{A}(1 - t(x)) \tag{1}$$

where $\mathbf{I}$ indicates the foggy image, $\mathbf{O}$ stands for the restored image, $\mathbf{A}$ represents the global atmospheric light, $t(x)$ denotes the fog transmission map, and x means the spatial location of each pixel. The transmission map is calculated as

$$t(x) = e^{(-\beta d(x))} \tag{2}$$

where β means the attenuation coefficient of the atmosphere and d measures the distance between land cover and sensor.

[1] https://pan.baidu.com/s/1oA7VdoSQYpOYsaeU2X2JDw(Download code:4696)

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:54:16 UTC from IEEE Xplore. Restrictions apply.

Second, the dark-object subtraction model is shown as

$$\mathbf{I}(x) = \mathbf{O}(x) + \mathbf{F}(x) \tag{3}$$

where $\mathbf{I}$ indicates the observed image acquired by imaging system, $\mathbf{O}$ represents the clear image need to be reconstructed, and $\mathbf{F}$ denotes the fog component. The desired clear image is calculated by subtracting the fog component $\mathbf{F}$ from the observed image $\mathbf{I}$.

Atmospheric scattering models and their improved versions have achieved satisfactory dehazing performance for RGB or multispectral images. However, they are designed to remove haze from RGB images or multispectral images, which cannot be directly applied for hyperspectral image processing. The main reason is that these atmospheric scattering models are based on an assumption that the attenuation coefficient for all spectral channels is a constant. The influence of different wavelengths for attenuation coefficient is not taken into consideration, which leads to an inaccurate haze transmission map for each spectral channel. Thus, the haze in original images cannot well removed.

III. Proposed Method

The schematic of the proposed method is presented in Fig. 2, which mainly contains two key steps: construction of fog model and image defogging based on fog model.

A. Fog Model

In this work, the foggy hyperspectral image is modeled as a superimposition of clean image and foggy image. Our goal is to estimate the fog intensity in different bands. Specifically, the fog model is expressed as

$$\mathbf{I}(x) = \mathbf{O}(x) + \mathbf{A}(x)\mathbf{F}(x) \tag{4}$$

where $\mathbf{I}$ denotes the foggy image, $\mathbf{O}$ is the clean image, $\mathbf{A}$ represents the fog abundance in different bands, $\mathbf{F}$ stands for the fog intensity map, and x denotes the position of the pixel in the original HSI. The term $\mathbf{A}(x)\mathbf{F}(x)$ indicates the fog intensity in different spectral channels.

B. Image Defogging Based on Fog Model

Hyperspectral defogging is to restore the pixel reflectance $\mathbf{O}$ from the foggy data $\mathbf{I}$. Accordingly, the fog abundance in different bands and the fog intensity map needs to be calculated according to the proposed fog model.

1) Estimation of Fog Intensity Map: Hyperspectral image consists of hundreds of spectral bands, and different wavelengths have different abilities to penetrate the fog. Wang and Yuan [42] and Feng *et al.* [43] suggested that the near-infrared bands have stronger penetration ability over visible bands since the wavelength in the near-infrared bands is longer than visible bands and is thus less scattered by the atmospheric particles. Fig. 3 shows the influence of fog in different bands. The bands falling into visible wavelength (0.36–0.65 μm) are seriously contaminated with the fog, whereas the bands falling into infrared wavelength (2.01–2.36 μm) are insensitive to the fog. Based on this observation, to obtain the fog intensity map,

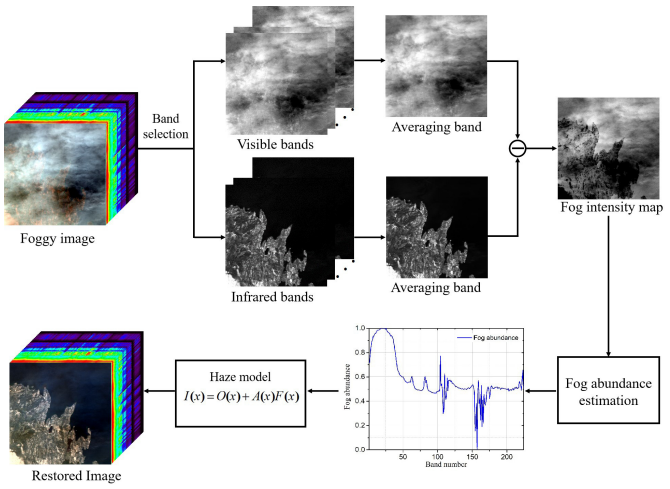


Fig. 2. Flowchart of the proposed method.

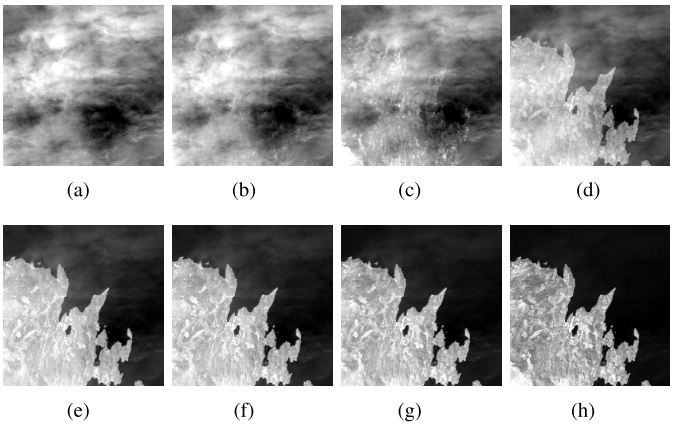


Fig. 3. Penetration of different bands to fog on AVIRIS data. (a) Band 10. (b) Band 20. (c) Band 30. (d) Band 40. (e) Band 50. (f) Band 60. (g) Band 80. (h) Band 100.

the foggy hyperspectral image $\mathbf{I}$ is first segmented into two groups, i.e., visible bands $\mathbf{I}_1$ (0.36–0.65 μm) and infrared bands $\mathbf{I}_2$ (2.01–2.36 μm). Then, the averaging-based operation is performed on the visible bands and the infrared bands

$$\hat{\mathbf{I}}_k = \frac{\sum_{a=1}^{N_{I_k}} \mathbf{I}_k^a}{N_{I_k}}, \quad k = 1, 2 \tag{5}$$

where $\hat{\mathbf{I}}_k$ is the averaging visible or infrared band and $N_{I_k}, with k = 1$ and 2 the total number of visible or infrared bands.

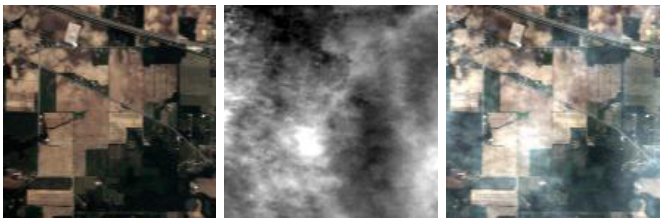
Finally, the fog intensity map is calculated by subtracting the averaging infrared band from the averaging visible band

$$\mathbf{F} = \hat{\mathbf{I}}_1 - \hat{\mathbf{I}}_2 \tag{6}$$

where $\mathbf{F}$ is the fog intensity map.

2) Estimation of Fog Abundance: The fog intensity map not only contains the fog but also has the details of land covers. In addition, the fog intensity of different spectral bands is different in HSI. Thus, we need to estimate the fog abundance in different spectral bands so as to obtain the fog intensity of different channels. More concretely, two pixels belonging

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:54:16 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c)

Fig. 4. Simulated dataset. (a) False color composite. (b) Fog mask. (c) Three bands of simulated dataset.

Algorithm 1 Fog Model-Based Hyperspectral Image Defogging

Input: Foggy image $\mathbf{I}$

Initialize (x_1, y_1), (x_2, y_2)
Construct fog model via Eq. (4)
Compute the fog intensity map $\mathbf{F}$ via Eq. (6)
Compute the fog abundance $\mathbf{A}$ via Eq. (7)
Compute the defogging result $\mathbf{O}$ via Eq. (8)

Output: Clean image $\mathbf{O}$

to the same object, i.e., (x_1, y_1) and (x_2, y_2), are selected manually from the fog intensity map so as to estimate the fog abundance, in which one pixel is from the object corrupted by fog and another is from the clean object. Then, the fog abundance is estimated as follows:

$$\mathbf{A} = \frac{\mathbf{I}_i - \mathbf{I}_j}{\mathbf{F}_i - \mathbf{F}_j} \tag{7}$$

where $\mathbf{A}$ is the fog abundance in different channels, $\mathbf{I}_i$ and $\mathbf{I}_j$ denote the spectral reflectance of the same land cover at ith and jth pixels, respectively, and $\mathbf{F}_i$ and $\mathbf{F}_j$ are the fog intensity value of ith and jth pixels, respectively.

3) Solving Haze Model: When the fog intensity map and fog abundance are obtained, the clean hyperspectral image can be easily obtained as follows:

$$\mathbf{O} = \mathbf{I} - \mathbf{AF} \tag{8}$$

where $\mathbf{O}$ is the defogging result.

IV. Experiments

A. Dataset

In this work, a benchmark foggy hyperspectral dataset is constructed, which is collected from the AVIRIS sensor and the China GF-5 satellite. This dataset consists of 40 images. The detailed descriptions are given as follows.

1) Simulated Dataset: To quantitatively measure the defogging performance of different approaches, a public dataset, i.e., Indian Pines, from the AVIRIS sensor is utilized to construct the simulated foggy dataset. To be specific, a foggy intensity map is first extracted from the real foggy dataset. Then, the foggy intensity map is taken as a mask and added into a clear dataset according to the proposed model (4). The clear dataset is composed of 200 spectral channels from 0.4 to 2.5 μm and has 145×145 pixels. Fig. 4 shows the simulated dataset and fog mask.

TABLE I
Some Details of the AVIRIS Dataset

Images	Captured place	Resolution	Fight time
A01	Rocky mountains	15.2m	8/8/2019
A02	Missoula	15.5m	8/8/2019
A03	Arizona	15.5m	8/21/2019
A04	Arizona	15.5m	8/21/2019
A05	Arizona	15.5m	8/21/2019
A06	Arizona	15.5m	8/21/2019
A07	Arizona	15.5m	8/21/2019
A08	Rocky mountains	15.2m	8/8/2019
A09	Juniper mountains	15.1m	8/21/2019
A10	Juniper mountains	15.1m	8/21/2019
A11	Juniper mountains	15.1m	8/21/2019
A12	Juniper mountains	15.1m	8/21/2019
A13	Juniper mountains	15.1m	8/21/2019
A14	Rocky mountains	15.2m	8/8/2019
A15	Rocky mountains	15.2m	8/8/2019
A16	Rocky mountains	15.2m	8/8/2019
A17	Rocky mountains	15.2m	8/8/2019
A18	Rocky mountains	15.2m	8/8/2019
A19	Rocky mountains	15.2m	8/8/2019
A20	Wallowa mountains	15.2m	8/6/2019

2) Real Datasets: The following conditions hold.

a) AVIRIS sensor: The AVIRIS sensor is embedded in airplane. This sensor can record 224 contiguous spectral bands with wavelengths from 0.4 to 2.5 μm. All images are 512×512 pixels, which are obtained by cropping large image downloaded from the AVIRIS website. Due to different flight heights, the spatial resolution of different images is different. The related information about the constructed database is shown in Table I. The three-band composite images are presented in the first column of Figs. 7 and 8. It is easy to observe that these images are contaminated with different levels of fog, and thus, the details of some land covers cannot be clearly displayed.

b) China GF-5 satellite: The hyperspectral images were captured by the Advanced Hyperspectral Imager (AHSI) carried on the China GF-5 satellite. The AHSI can provide visible/near-infrared and shortwave infrared data. This image consists of 330 spectral bands with wavelength range varying from 0.4 to 2.5 μm. The spatial resolution is 30 m/pixel. Due to the serious water and atmospheric absorption, 25 bands (No. 193–200 and 246–262) are removed before defogging. The three-band composite images are presented in the first column of Figs. 9 and 10.

B. Compared Methods

To examine the advantage of the proposed defogging method, several dehazing methods in the computer vision field and remote sensing field are used as competitors, including DCP-based method [13], multiscale convolutional neural network (MSCNN) method [37], DHIM-based remote sensing dehazing method [29], multilayer perceptron (MLP)-based single-image dehazing method [38], IDM for remote sensing image [33], adaptive dehazing method (ADM) for single remote sensing image [31], and radiance–reflectance combined optimization (RRCO) method [44]. These approaches are selected as competitors since they are highly cited papers or

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:54:16 UTC from IEEE Xplore. Restrictions apply.

recently proposed methods. All parameters are set according to the authors' suggestions in the related publications.

C. Experiments on Simulated Dataset

Fig. 5 shows the defogging results of different approaches on the simulated dataset. As depicted in this figure, the seven compared approaches cannot well work in removing the fog from the corrupted image. The main reason is that these atmospheric scattering model-based approaches are not suitable for hyperspectral defogging. In contrast, it can be clearly observed that the proposed method can significantly remove the fog from the simulated dataset and improve the visibility of the contaminated image. The resulting image obtained by the proposed method has rich spatial information and color fidelity. Besides, Fig. 6 shows the spectral reflectance of two land covers, i.e., Soybean and Trees. It is obvious that the spectral curves obtained by the proposed method are more closer to the clear spectral curve of the same land cover, which also validates that the proposed method can well preserve the spectral information of different types of objects.

To quantitatively assess the defogging performance of different approaches, five widely used objective metrics are adopted, including spectral angle mapper (SAM) [45], structural similarity (SSIM) [46], peak signal-to-noise ratio (PSNR) [47], visual image quality (VIQ) [48], and universal quality index (UQI) [49]. All quality indexes are estimated by using the clean original image as a reference.

1) SAM: It measures the spectral similarity between the restored and referenced data

$$\mathrm{SAM}\left(X,\hat{X}\right)=\frac{1}{N}\sum_{i=1}^{N}\arccos\frac{\hat{X}_j^T X_j}{\left\|\hat{X}_j\right\|_2\left\|X_j\right\|_2} \tag{9}$$

where X is the original image, $\hat{X}$ is the reconstructed image, and N denotes the number of spectral channels.

2) SSIM: It calculates the structural similarities between the restored and referenced data

$$\mathrm{SSIM}\left(X,\hat{X}\right)=\frac{1}{N}\sum_{i=1}^{N}\frac{\left(2\tilde{\mu}_{X_i}\tilde{\mu}_{\hat{X}_i}+c_1\right)\left(2\sigma_{X_i\hat{X}_i}+c_2\right)}{\left(\tilde{\mu}_{X_i}^2+\tilde{\mu}_{\hat{X}_i}^2+c_1\right)\left(\sigma_{\hat{X}_i}^2+\sigma_{\hat{X}_i}^2+c_2\right)} \tag{10}$$

where N is the total number of spectral channels, X_i is the ith band of X, $\tilde{\mu}_{X_i}$ and $\tilde{\mu}_{\hat{X}_i}$ are the average of X_i and $\hat{X}_i$, respectively, $\sigma_{X_i}^2$ and $\sigma_{\hat{X}_i}^2$ are the variances of X_i and $\hat{X}_i$, respectively, and $\sigma_{X_i\hat{X}_i}$ is the covariance of X_i and $\hat{X}_i$.

3) PSNR: It denotes the spatial reconstruction quality between the restored and referenced data

$$\mathrm{PSNR}\left(X,\hat{X}\right)=\frac{1}{N}10\log_{10}\left(\frac{(\max(X_i))^2}{\left\|X_i-\hat{X}_i\right\|_2^2/PQ}\right) \tag{11}$$

where PQ is the spatial dimension of the input image.

4) VIF: It indicates the mutual information between the restored and referenced data

$$VIF\left(X,\hat{X}\right)=\frac{\sum_{k=1}^{K}\left[I\left(C;\hat{X}|z\right)\right]}{\sum_{k=1}^{K}\left[I\left(C;X|z\right)\right]} \tag{12}$$

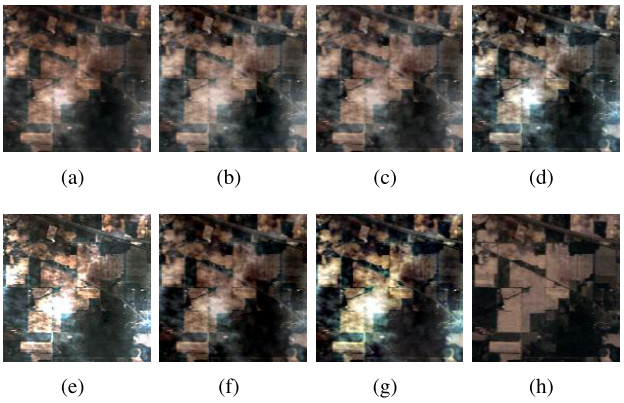
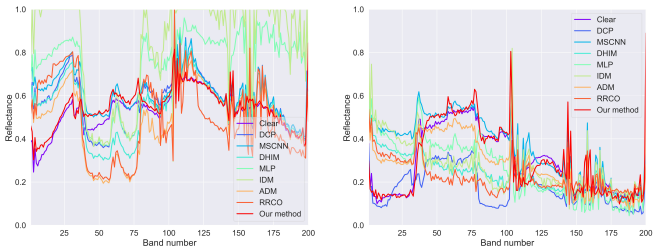


Fig. 5. Resulting images of different methods on simulated dataset. (a) DCP [13]. (b) MSCNN [37]. (c) DHIM [29]. (d) MLP [38]. (e) IDM [33]. (f) ADM [31]. (g) RRCO [44]. (h) Our method.

TABLE II

QUANTITATIVE COMPARISON OF ALL CONSIDERED APPROACHES ON SIMULATED FOGGY DATASET. THE BEST PERFORMANCE IS HIGHLIGHTED WITH BOLD

Methods	SAM	SSIM	PSNR	VIF	QUI
Best value	0	1	$+\infty$	1	1
DCP	0.3377	0.6762	14.6295	0.5313	0.6255
MSCNN	0.1545	0.9085	25.8365	0.7016	0.8797
DHIM	0.2262	0.8558	21.9121	0.7519	0.8264
MLP	0.2832	0.7205	16.2691	0.6504	0.6949
IDM	0.3437	0.6133	14.1409	0.4564	0.5959
ADM	0.1404	0.9039	23.2267	0.6801	0.8724
RRCO	0.3162	0.8019	17.9528	0.6338	0.7731
Our method	**0.0816**	**0.9577**	**32.0969**	**0.7576**	**0.9428**



(a) (b)

Fig. 6. Spectral reflectance of land covers. (a) Soybean (88, 54). (b) Trees (106, 63).

where $I(C;\hat{X}|z)$ denotes the mutual information between C and $\hat{X}$ and $I(C;X|z)$ is the mutual information between C and X.

5) UQI: It stands for the similarity on luminance, contrast, and structure between the restored and referenced data

$$\mathrm{UQI}\left(X,\hat{X}\right)=\frac{1}{N}\sum_{i=1}^{N}\frac{4\sigma_{X_i\hat{X}_i}\mu_{X_i}\mu_{\hat{X}_i}}{\left(\sigma_{X_i}^2+\sigma_{\hat{X}_i}^2\right)\left(\mu_{X_i}^2+\mu_{\hat{X}_i}^2\right)} \tag{13}$$

where $\sigma_{X_i\hat{X}_i}$ is the covariance of X_i and $\hat{X}_i$ and μ_{X_i} and $\mu_{\hat{X}_i}$ are the mean values of X_i and $\hat{X}_i$, respectively.

Table II shows the quantitative comparison of all considered approaches. The objective results in Table II illustrate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:54:16 UTC from IEEE Xplore. Restrictions apply.

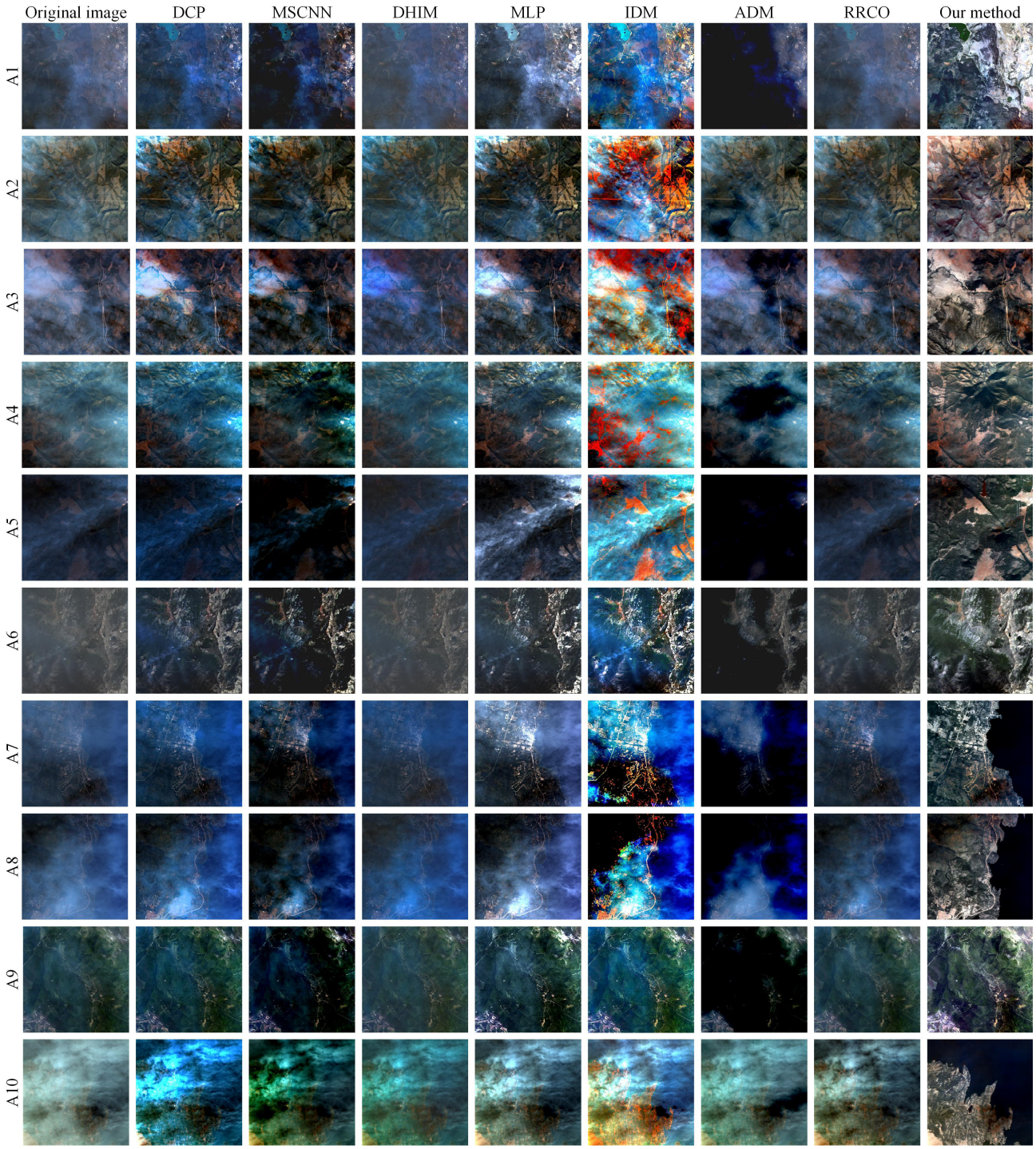


Fig. 7. Defogging results of different methods on the AVIRIS dataset (A1–A10).

that our method can consistently outperform other defogging approaches in terms of all metrics, which is consistent with visual results. In contrast with representative DCP method, the proposed method improves SAM from 0.3377 to 0.0816, SSIM from 0.6762 to 0.9577, PSNR from 14.6295 to 32.0969, VIF from 0.5313 to 0.7576, and QUI from 0.6255 to 0.9428. These results verify that the proposed method can better preserve the spectral reflectance of different objects and restore high-quality images.

D. Experiments on Real Datasets

1) AVIRIS Dataset: Figs. 7 and 8 show the defogging results of different methods on the AVIRIS dataset. The visual results of three bands (No. 12, 19, and 29) are displayed. The DCP method was originally proposed for RGB images. The resulting images obtained by the DCP method are in low contrast. The MSCNN method is slightly better than the DCP method, but the fog in the original images still cannot be removed completely. The reason is that the MSCNN trained on the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:54:16 UTC from IEEE Xplore. Restrictions apply.

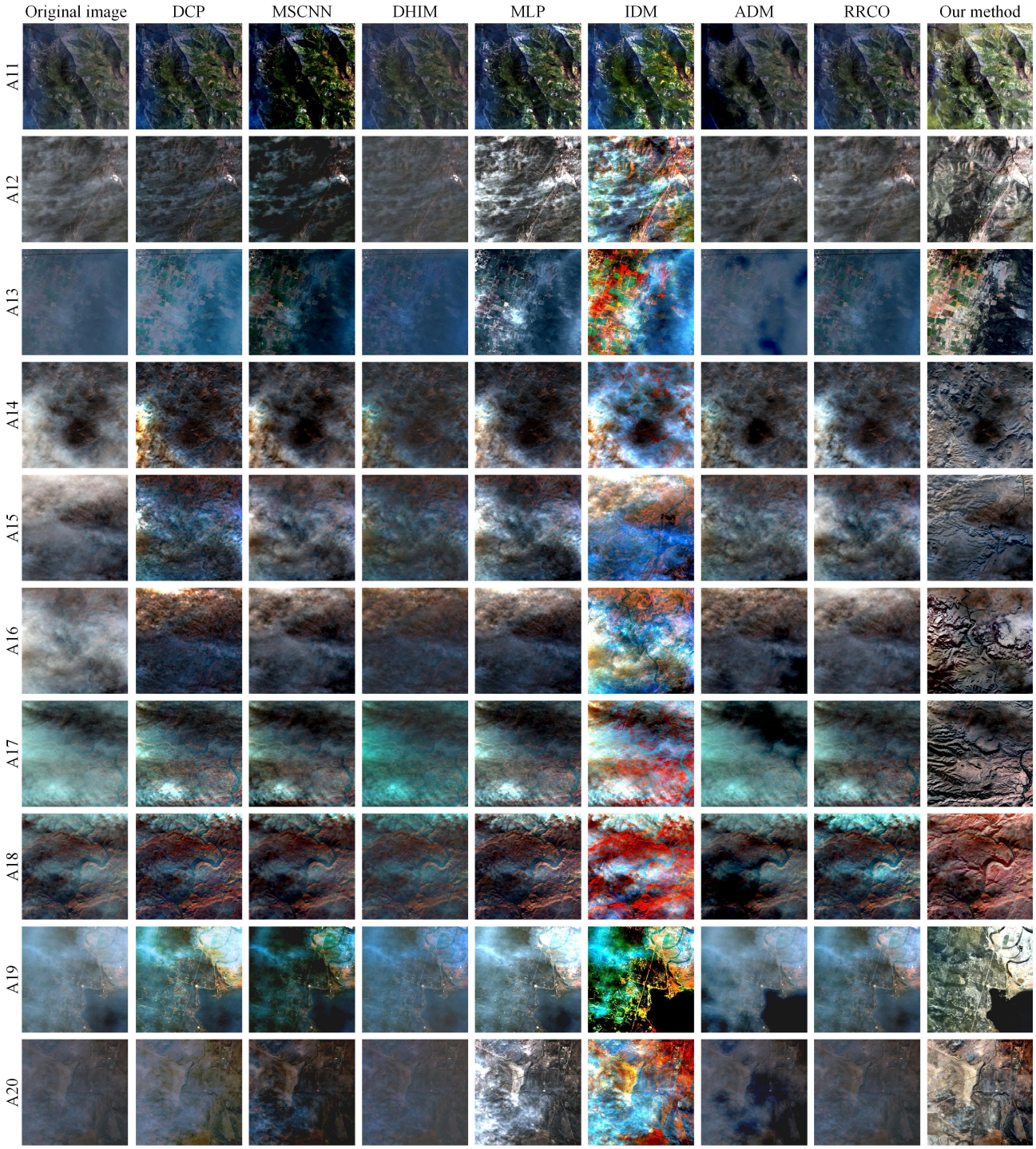


Fig. 8. Defogging results of different methods on the AVIRIS dataset (A11–A20).

synthesized hazy images fails to yield the mapping between the fog images and the clean images. The DHIM method is an improved version of DCP for multispectral images, which also cannot work in removing fog from hyperspectral images. The MLP method increases the brightness of the resulting images since a contrast stretching technique is used to improve the dynamic range of the restored images. For the IDM method, the resulting images are higher brightness over other methods. Nevertheless, the colors in the resulting images appear obvious distortions. The ADM method is failed for the fog removal in hyperspectral images since the dark channel-saturation prior cannot build the correlation between dark channel and saturation of haze-free images. The RRCO method also cannot remove the dense fog. In contrast, our method can better restore the spatial details of land covers than other compared methods.

2) GF Dataset: Figs. 9 and 10 show the defogging results of different approaches on the GF dataset. The results of three

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:54:16 UTC from IEEE Xplore. Restrictions apply.

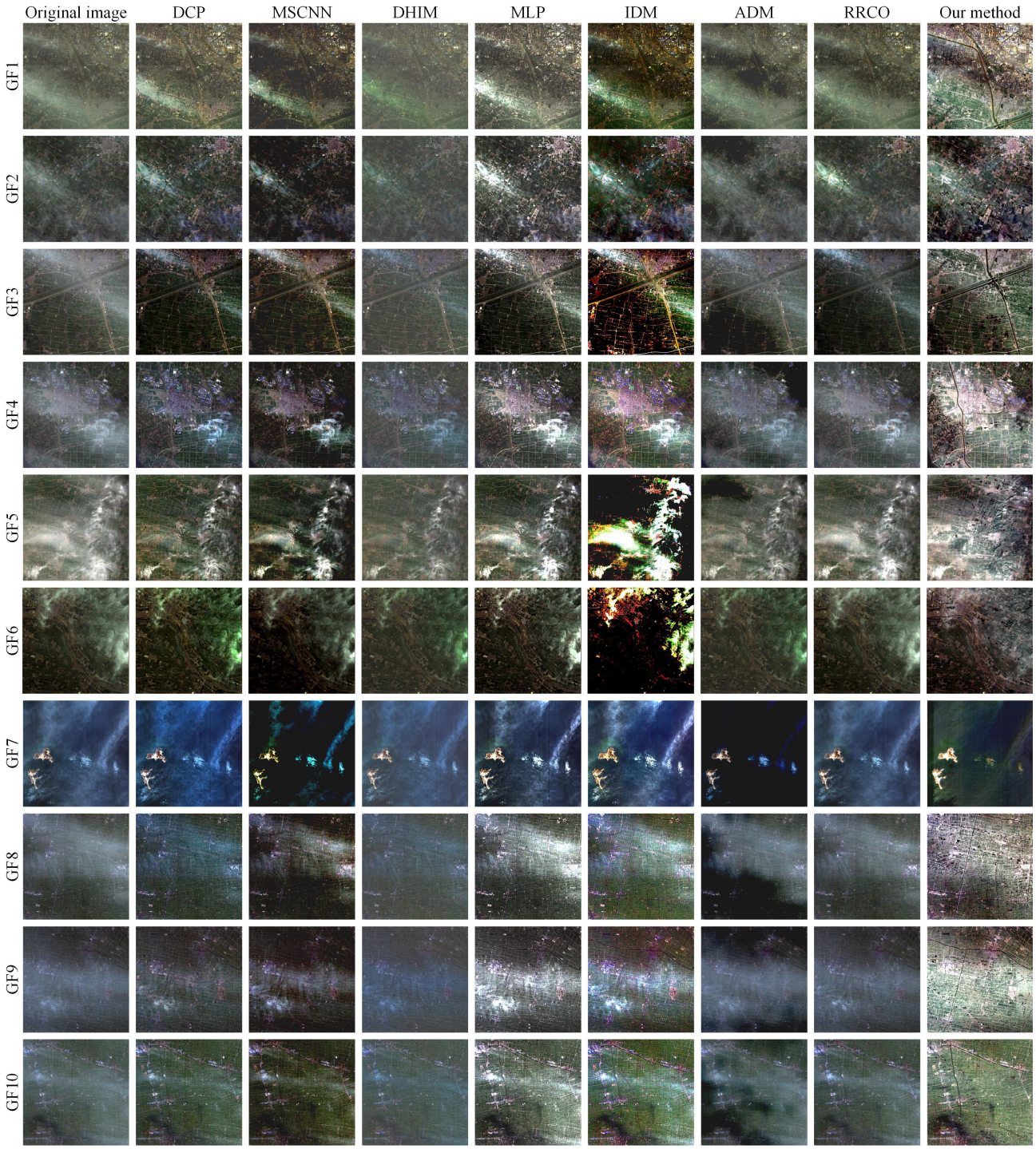


Fig. 9. Defogging results of different methods on the GF dataset (GF1–GF10).

bands (No. 58, 38, and 20) are presented. The original images are contaminated with different levels of fog. From Figs. 9 and 10, several phenomenons can be easily observed. First, most existing methods cannot be directly applied for defogging in hyperspectral images, since they usually produce unsatisfactory performance (see the second column to eighth column in Figs. 9 and 10). This is due to the fact that these methods are designed to tackle RGB or multispectral data. Second, the MSCNN and IDM methods can enhance the clarity of defogging results. The reason is that contrast enhancement is adopted in these methods. Third, the classical DCP and its improved versions are not suitable for achieving defogging from hyperspectral images. Finally, our method can obtain the best defogging performance compared to other methods. Our method effectively removes the fog in the original images and also preserves the color fidelity.

Furthermore, we also validate the defogging performance of the proposed method on a large-scale GF dataset. Fig. 11

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:54:16 UTC from IEEE Xplore. Restrictions apply.

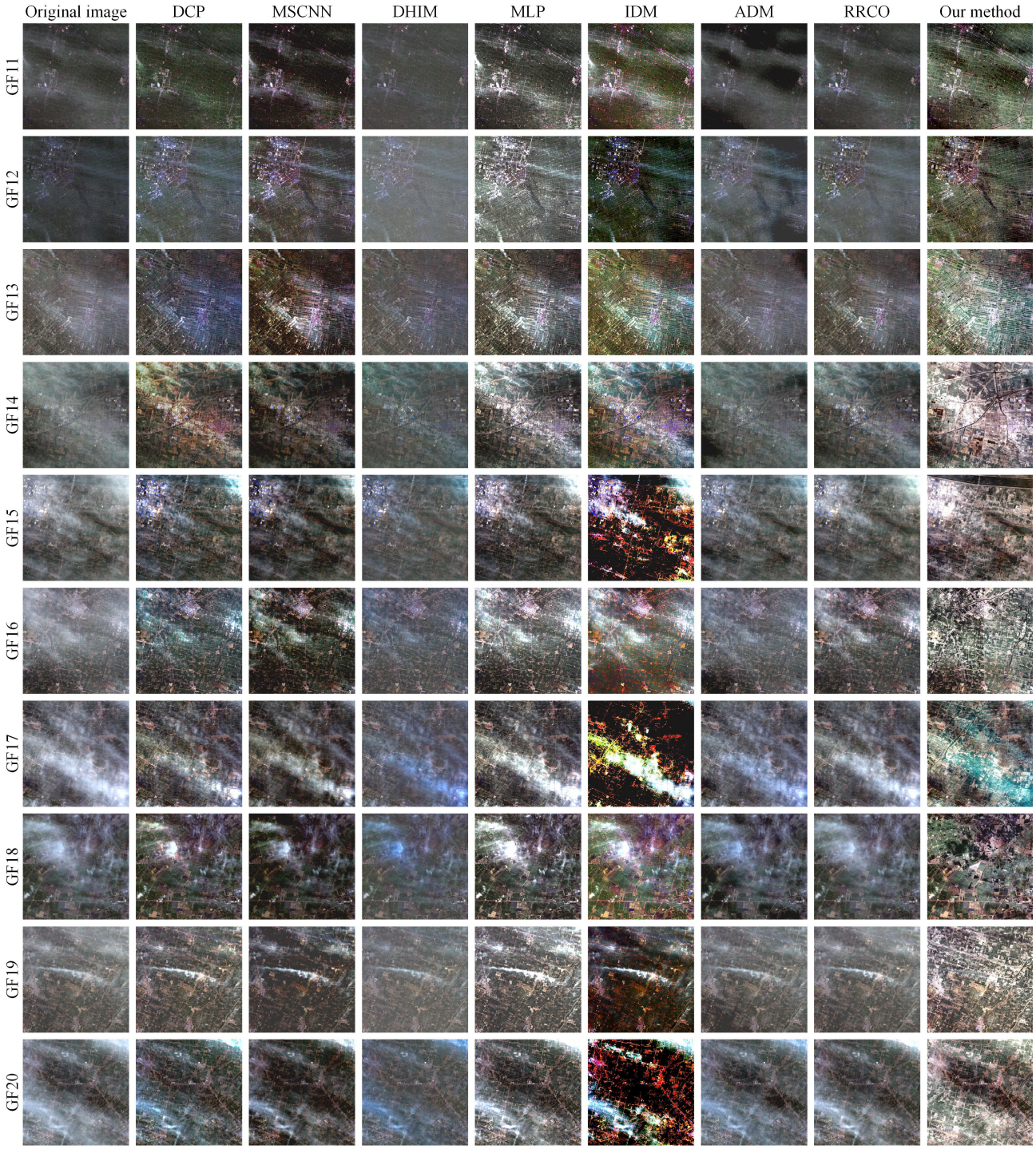
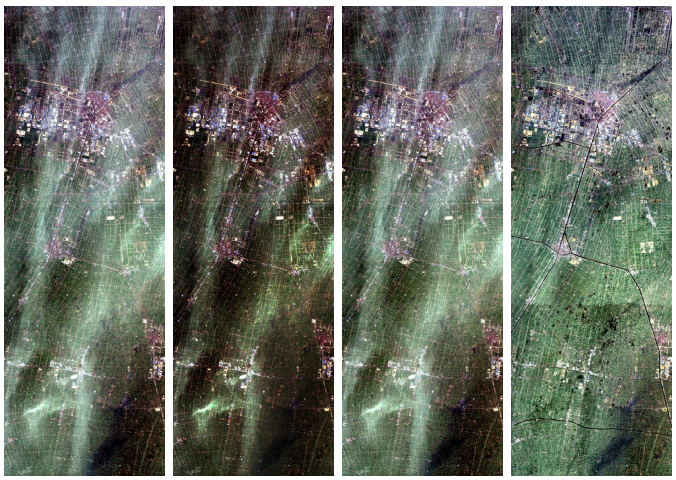


Fig. 10. Defogging results of different methods on the GF dataset (GF11–GF20).

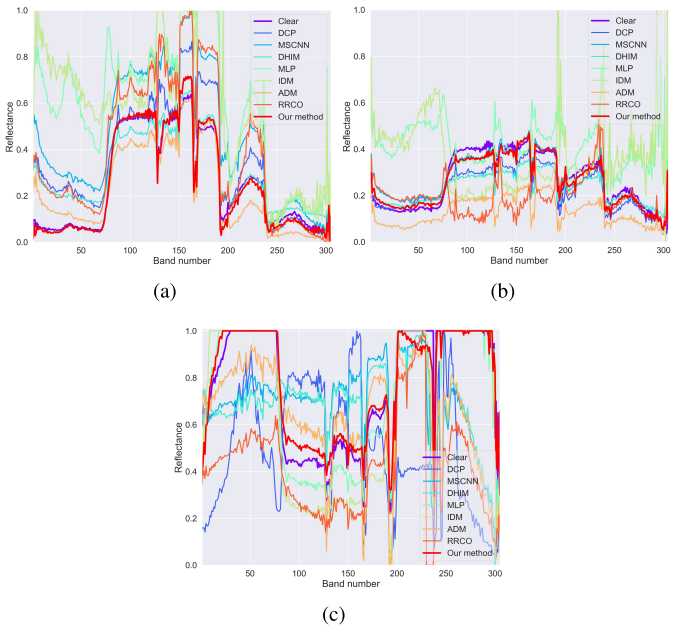
shows the defogging results of several popular techniques. In this figure, the DCP and IDM approaches boost the contrast of the original image. Nevertheless, the thick fog in the original image cannot be removed effectively since these models aim at removing haze from visible and multispectral images. Based on the visual comparison, it is obvious that the proposed method can yield much clearer dehazed results with finer details and fewer artifacts with respect to other approaches. Moreover, Fig. 12 shows the spectral reflectance of different approaches on the large-scale GF dataset. Here, the position of clean plant is located at (177, 589). The coordinate of the clean road is located at (57, 581), and the position of clean roof is located at (507, 375). It is found from Fig. 12 that the spectral curves of the proposed method are more similar to the spectral reflectance of the clean land covers.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:54:16 UTC from IEEE Xplore. Restrictions apply.



(a) (b) (c) (d)

Fig. 11. Defogging results of different methods on large-scale GF dataset. (a) Original image. (b) DCP method [13]. (c) IDM method [33]. (d) Proposed method.



(a) (b)

(c)

Fig. 12. Spectral reflectance of different land covers. (a) Plant (160, 212). (b) Road (299, 574). (c) Roof (150, 362).

E. Computing Time

Finally, the execution time of all considered techniques is shown in Fig. 13. Since all sample images are the same spatial size, only two images, i.e., A1 and GF1, are adopted to compare the running time of different approaches. All approaches are conducted on a Laptop that is equipped with 2.6 GHz and 8-GB RAM using MATLAB 2014a. It can be observed from Fig. 13 that the execution time of our method is very fast among all considered approaches, which is more suitable for practical applications. The reason is that our method only requires dot division and dot multiplication.

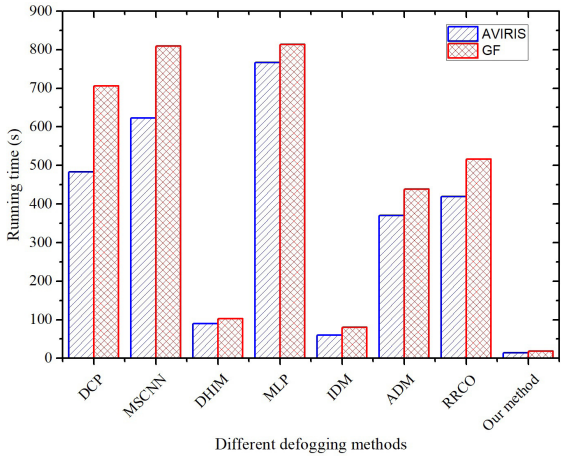


Fig. 13. Running time of different methods.

V. Conclusion

Hyperspectral image fog removal is a challenging problem. Current dehazing approaches in the computer vision field or multispectral remote sensing field are not well suitable for fog removal from hyperspectral images. In this work, a novel model-based defogging method is proposed for hyperspectral images. First, we create a publicly available benchmark database for defogging from hyperspectral images. This database is composed of 40 images obtained from airborne and spaceborne sensors. Then, a fog model-based method is developed for defogging of hyperspectral images. Finally, the effectiveness of the proposed defogging model is validated on the constructed database. Experimental results reveal that the proposed method can achieve favorable performance with respect to other state-of-the-art dehazing methods.

Acknowledgment

The authors would like to thank the Editor-in-Chief, the anonymous Associate Editor, and the reviewers for their insightful comments and suggestions, which have greatly improved the quality of this article.

References

[1] G. Cheng, Z. Li, J. Han, X. Yao, and L. Guo, "Exploring hierarchical convolutional features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6712–6722, Nov. 2018.

[2] G. Cheng, C. Yang, X. Yao, L. Guo, and J. Han, "When deep learning meets metric learning: Remote sensing image scene classification via learning discriminative CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2811–2821, May 2018.

[3] Y. Gu *et al.*, "Multimodal hyperspectral remote sensing: An overview and perspective," *Sci. China Inf. Sci.*, vol. 64, no. 2, pp. 1869–1919, Feb. 2021.

[4] P. Duan, P. Ghamisi, X. Kang, B. Rasti, S. Li, and R. Gloaguen, "Fusion of dual spatial information for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, early access, Nov. 11, 2020, doi: 10.1109/TGRS.2020.3031928.

[5] G. Gao and Y. Gu, "Tensorized principal component alignment: A unified framework for multimodal high-resolution images classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 46–61, Jan. 2019.

[6] X. Kang, X. Xiang, S. Li, and J. A. Benediktsson, "PCA-based edge-preserving features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 12, pp. 7140–7151, Dec. 2017.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:54:16 UTC from IEEE Xplore. Restrictions apply.

[7] X. Kang, S. Li, L. Fang, and J. A. Benediktsson, "Intrinsic image decomposition for feature extraction of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 4, pp. 2241–2253, Apr. 2015.
[8] C. He, B. Gao, Q. Huang, Q. Ma, and Y. Dou, "Environmental degradation in the urban areas of China: Evidence from multi-source remote sensing data," *Remote Sens. Environ.*, vol. 193, pp. 65–75, May 2017.
[9] P. Duan, J. Lai, J. Kang, X. Kang, P. Ghamisi, and S. Li, "Texture-aware total variation-based removal of sun glint in hyperspectral images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 359–372, Aug. 2020.
[10] K. Berger *et al.*, "Crop nitrogen monitoring: Recent progress and principal developments in the context of imaging spectroscopy missions," *Remote Sens. Environ.*, vol. 242, Jun. 2020, Art. no. 111758.
[11] M. Kanning, B. Siegmann, and T. Jarmer, "Regionalization of uncovered agricultural soils based on organic carbon and soil texture estimations," *Remote Sens.*, vol. 8, no. 11, p. 927, Nov. 2016.
[12] P. Duan, X. Kang, S. Li, P. Ghamisi, and J. A. Benediktsson, "Fusion of multiple edge-preserving operations for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 10336–10349, Dec. 2019
[13] K. He, J. Sun, and X. Tang, "Single image haze removal using dark channel prior," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 12, pp. 2341–2353, Dec. 2011.
[14] B. Cai, X. Xu, K. Jia, C. Qing, and D. Tao, "DehazeNet: An end-to-end system for single image haze removal," *IEEE Trans. Image Process.*, vol. 25, no. 11, pp. 5187–5198, Nov. 2016.
[15] A. Wang, W. Wang, J. Liu, and N. Gu, "AIPNet: Image-to-image single image dehazing with atmospheric illumination prior," *IEEE Trans. Image Process.*, vol. 28, no. 1, pp. 381–393, Jan. 2019.
[16] Q. Liu, X. Gao, L. He, and W. Lu, "Single image dehazing with depth-aware non-local total variation regularization," *IEEE Trans. Image Process.*, vol. 27, no. 10, pp. 5178–5191, Oct. 2018.
[17] Q. Zhu, J. Mai, and L. Shao, "A fast single image haze removal algorithm using color attenuation prior," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 3522–3533, Nov. 2015.
[18] R. T. Tan, "Visibility in bad weather from a single image," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2008, pp. 1–8.
[19] C.-Y. Li, J.-C. Guo, R.-M. Cong, Y.-W. Pang, and B. Wang, "Underwater image enhancement by dehazing with minimum information loss and histogram distribution prior," *IEEE Trans. Image Process.*, vol. 25, no. 12, pp. 5664–5677, Dec. 2016.
[20] C. O. Ancuti and C. Ancuti, "Single image dehazing by multi-scale fusion," *IEEE Trans. Image Process.*, vol. 22, no. 8, pp. 3271–3282, Aug. 2013.
[21] R. Richter, "Atmospheric correction of satellite data with haze removal including a haze/clear transition region," *Comput. Geosci.*, vol. 22, no. 6, pp. 675–681, Jul. 1996.
[22] J. H. Jang, S. D. Kim, and J. B. Ra, "Enhancement of optical remote sensing images by subband-decomposed multiscale retinex with hybrid intensity transfer function," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 5, pp. 983–987, Sep. 2011.
[23] C. Feng, J.-W. Ma, Q. Dai, and X. Chen, "An improved method for cloud removal in ASTER data change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, vol. 5, Sep. 2004, pp. 3387–3389.
[24] Y. Du, B. Guindon, and J. Cihlar, "Haze detection and removal in high resolution satellite image with wavelet analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 1, pp. 210–217, Jan. 2002.
[25] J. Li, Q. Hu, and M. Ai, "Haze and thin cloud removal via sphere model improved dark channel prior," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 3, pp. 472–476, Mar. 2019.
[26] A. Makarau, R. Richter, R. Müller, and P. Reinartz, "Haze detection and removal in remotely sensed multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 9, pp. 5895–5905, Sep. 2014.
[27] D. Singh and V. Kumar, "Dehazing of remote sensing images using improved restoration model based dark channel prior," *Imag. Sci. J.*, vol. 65, no. 5, pp. 282–292, 2017.
[28] D. Singh and V. Kumar, "Dehazing of remote sensing images using fourth-order partial differential equations based trilateral filter," *IET Comput. Vis.*, vol. 12, no. 2, pp. 208–219, Mar. 2018.
[29] X. Pan, F. Xie, Z. Jiang, and J. Yin, "Haze removal for a single remote sensing image based on deformed haze imaging model," *IEEE Signal Process. Lett.*, vol. 22, no. 10, pp. 1806–1810, Oct. 2015.
[30] J. Long, Z. Shi, W. Tang, and C. Zhang, "Single remote sensing image dehazing," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 59–63, Jan. 2014.
[31] F. Xie, J. Chen, X. Pan, and Z. Jiang, "Adaptive haze removal for single remote sensing image," *IEEE Access*, vol. 6, pp. 67982–67991, 2018.
[32] Q. Guo, H.-M. Hu, and B. Li, "Haze and thin cloud removal using elliptical boundary prior for remote sensing image," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9124–9137, Nov. 2019.
[33] L. Xu, D. Zhao, Y. Yan, S. Kwong, J. Chen, and L.-Y. Duan, "IDeRs: Iterative dehazing method for single remote sensing image," *Inf. Sci.*, vol. 489, pp. 50–62, Jul. 2019.
[34] H. Jiang, N. Lu, L. Yao, and X. Zhang, "Single image dehazing for visible remote sensing based on tagged haze thickness maps," *Remote Sens. Lett.*, vol. 9, no. 7, pp. 627–635, Jul. 2018.
[35] M. Qin, F. Xie, W. Li, Z. Shi, and H. Zhang, "Dehazing for multispectral remote sensing images based on a convolutional neural network with the residual architecture," *IEEE J. Sel. Topics Appl. Earth Observ.*, vol. 11, no. 5, pp. 1645–1655, May 2018.
[36] J. Guo, J. Yang, H. Yue, H. Tan, C. Hou, and K. Li, "RSDehazeNet: Dehazing network with channel refinement for multispectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no 3, pp. 1–15, Mar. 2020.
[37] W. Ren, S. Liu, H. Zhang, J. Pan, X. Cao, and M.-H. Yang, "Single image dehazing via multi-scale convolutional neural networks," in *Proc. Comput. Vis. ECCV*, Sep. 2016, pp. 154–169.
[38] S. Salazar-Colores, I. Cruz-Aceves, and J. M. Ramos-Arreguin, "Single image dehazing using a multilayer perceptron," *J. Electron. Imag.*, vol. 27, no. 4, pp. 1–11, Jul. 2018.
[39] P. S. Chavez, Jr., "An improved dark-object subtraction technique for atmospheric scattering correction of multispectral data," *Remote Sens. Environ.*, vol. 24, no. 3, pp. 459–479, Apr. 1988.
[40] S. Chen, X. Chen, J. Chen, P. Jia, X. Cao, and C. Liu, "An iterative haze optimized transformation for automatic cloud/haze detection of Landsat imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 5, pp. 2682–2694, May 2016.
[41] Y. Zhang, B. Guindon, and J. Cihlar, "An image transform to characterize and compensate for spatial variations in thin cloud contamination of Landsat images," *Remote Sens. Environ.*, vol. 82, nos. 2–3, pp. 173–187, 2002.
[42] W. Wang and X. Yuan, "Recent advances in image dehazing," *IEEE/CAA J. Autom. Sinica*, vol. 4, no. 3, pp. 410–436, Jul. 2017.
[43] C. Feng, S. Zhuo, X. Zhang, L. Shen, and S. Süsstrunk, "Near-infrared guided color image dehazing," in *Proc. IEEE Int. Conf. Image Process.*, Sep. 2013, pp. 2363–2367.
[44] J. Shin, M. Kim, J. Paik, and S. Lee, "Radiance-reflectance combined optimization and structure-guided ℓ_0-norm for single image dehazing," *IEEE Trans. Multimedia*, vol. 22, no. 1, pp. 30–44, Jan. 2020.
[45] P. Duan *et al.*, "Component decomposition-based hyperspectral resolution enhancement for mineral mapping," *Remote Sens.*, vol. 12, no. 18, p. 2903, Sep. 2020.
[46] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
[47] A. Tanchenko, "Visual-PSNR measure of image quality," *J. Vis. Commun. Image Represent.*, vol. 25, no. 5, pp. 874–878, 2014.
[48] H. R. Sheikh and A. C. Bovik, "Image information and visual quality," *IEEE Trans. Image Process.*, vol. 15, no. 2, pp. 430–444, Feb. 2006.
[49] Z. Wang and A. C. Bovik, "A universal image quality index," *IEEE Signal Process. Lett.*, vol. 9, no. 3, pp. 81–84, Mar. 2002.

Xudong Kang (Senior Member, IEEE) received the B.Sc. degree from Northeast University, Shenyang, China, in 2007, and the Ph.D. degree from Hunan University, Changsha, China, in 2015.

In 2015, he joined the College of Electrical Engineering, Hunan University. His research interests include hyperspectral feature extraction, image classification, image fusion, and anomaly detection.

Dr. Kang received the National Nature Science Award of China (Second class and Rank as third) and the Second Prize in the Student Paper Competition in IGARSS 2014. He was selected as the Best Reviewer for the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS and the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. He was an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING from 2018 to 2019. He serves as an Associate Editor for the IEEE GEOSCIENCE AND REMOTE SENSING LETTERS and the IEEE JOURNAL ON MINIATURIZATION FOR AIR AND SPACE SYSTEMS.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:54:16 UTC from IEEE Xplore. Restrictions apply.

Zhengyao Fei received the B.S. and M.S. degrees from Hunan University, Changsha, China, in 2018 and 2021, respectively.

Her research interests include hyperspectral image restoration and classification.

Puhong Duan (Student Member, IEEE) received the B.Sc. degree from Suzhou University, Suzhou, China, in 2014, and the M.S. degree from Hefei University of Technology, Hefei, China, in 2017. He is currently pursuing the Ph.D. degree with the Laboratory of Vision and Image Processing, Hunan University, Changsha, China.

From October 2019 to December 2019, he was a Visiting Ph.D. Student with the Faculty of Electrical Engineering and Computer Science, Technische Universität Berlin, Berlin, Germany. From January 2020 to October 2020, he was a Visiting Ph.D. Student with the Helmholtz-Zentrum Dresden-Rossendorf, Helmholtz Institute Freiberg for Resource Technology, Freiberg, Germany. His research interests include hyperspectral image classification, visualization, and image fusion. His research interests include hyperspectral image classification, multimodal fusion, and object detection.

Shutao Li (Fellow, IEEE) received the B.S., M.S., and Ph.D. degrees from Hunan University, Changsha, China, in 1995, 1997, and 2001, respectively, all in electrical engineering.

In 2001, he joined the College of Electrical and Information Engineering, Hunan University. From May 2001 to October 2001, he was a Research Associate with the Department of Computer Science, The Hong Kong University of Science and Technology, Hong Kong. From November 2002 to November 2003, he was a Post-Doctoral Fellow with the Royal Holloway College, University of London, London, U.K. From April 2005 to June 2005, he was a Visiting Professor with the Department of Computer Science, The Hong Kong University of Science and Technology. He is currently a Full Professor with the College of Electrical and Information Engineering, Hunan University. He has authored or coauthored over 200 refereed articles. His research interests include image processing, pattern recognition, and artificial intelligence.

Dr. Li is a member of the Editorial Board of *Information Fusion* and *Sensing and Imaging*. He received two 2nd-Grade State Scientific and Technological Progress Awards of China in 2004 and 2006. He is also an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and the IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 11:54:16 UTC from IEEE Xplore. Restrictions apply.